

Berat Erkan Elcelik

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Profile

- AI Engineer and M.Sc. student specializing in multi-agent systems, RAG pipelines, and production ML infrastructure. Experience building scalable AI products with LangGraph, real-time voice systems, and GPU-optimized inference servers. **Open to AI Engineer roles across Europe.**

Education

University of Bremen

M.Sc., Artificial Intelligence and Intelligent Systems

10/2024 – Present

Bremen, Germany

Ankara University

B.Sc., Computer Engineering (GPA: 3.76/4)

10/2018 – 08/2023

Ankara, Turkey

Work Experience

AI Engineer, PvFritz UG

Part-Time · Remote

03/2025 – Present

Germany

- Architected a **hierarchical multi-agent AI system** (Supervisor + 5 specialized agents) using LangGraph and MCP, enabling context-aware automation across a multi-tenant Django application with RBAC-based access control.
- Built **hybrid RAG pipeline** combining pgvector semantic search with Qwen3 cross-encoder reranking (85% scoring weight), achieving high-precision document retrieval across isolated tenant data.
- Deployed **multi-GPU inference infrastructure** (2× RTX 5070) with Triton Inference Server, dynamic batching, and OpenResty/Nginx routing—achieving **15–17 req/s**, **40% latency reduction**, and **0% error rate** across 27K+ requests.
- Developed **15+ tenant-aware tools** for document operations, workflow execution, and CRUD actions with structured output schemas, JSON-based command execution, and fallback parsing strategies.
- Implemented **real-time observability**: agent reasoning traces, GPU metrics, token usage tracking, and cost estimation dashboards using SigNoz.

Data Scientist, TAI – Turkish Aerospace Industries

CGF Project – Autonomous Aircraft Development

08/2023 – 10/2024

Ankara, Turkey

- Developed **RL agents for autonomous aircraft** (F-16, SU-27, HURJET) implementing DDPG and PPO from scratch, with custom reward shaping for stable convergence.
- Built **multi-agent dogfight training environments** with 4 concurrent agents and adversarial self-play.
- Applied **time-series anomaly detection** using Isolation Forest, LOF, and LSTM Autoencoders on flight telemetry data.

Data Scientist (Part-Time), TAI – Turkish Aerospace Industries

Working Student

12/2022 – 08/2023

Ankara, Turkey

- Performed anomaly detection and flight data analysis; designed reward functions for RL training stability.
- Built custom **TD3 reinforcement learning model** for aircraft control optimization.

Projects

Real-Time Voice Customer Agent · Python, LangGraph, LiveKit, Twilio

2025

- Built a **production voice AI system** for car dealership customer service with real-time STT/TTS, multi-agent orchestration, and Twilio telephony integration.
- Implemented **supervisor-worker architecture** using LangGraph with specialized agents for inventory lookup, appointment booking, and FAQ handling with human-in-the-loop escalation.
- Engineered audio pipeline with proper **μ -law encoding and downsampling** for telephony-grade quality.

Agent Playground · Python, Django, LangGraph, React, PostgreSQL/pgvector

2025

- Built **multi-agent platform** using LangGraph Functional API with supervisor pattern, async tool execution, and human-in-the-loop checkpointing for complex task orchestration.
- Implemented **RAG with semantic search** (pgvector), document processing with OCR, and Langfuse observability for cost tracking and performance monitoring.
- Developed React/TypeScript frontend with real-time streaming chat and Docker-based deployment.

RAG Reranker Evaluator · Python, Vectara RAGBench, BEIR

2025

- Built **evaluation framework** for benchmarking embedding models and cross-encoder rerankers using Vectara Open RAGBench dataset across multiple retrieval metrics (NDCG, MRR, Recall@K).
- Compared dense retrievers (BGE, E5) with reranking strategies to quantify **retrieval quality vs. latency tradeoffs**.

TheUnichat – Mobile App Startup · React Native, Supabase

2024 – 2025

- Founded and developed a **React Native** mobile app for university student communities; launched on iOS/Android reaching **500+ users** in the first week. Led design, development, and marketing.

Skills

- **Languages:** Python, JavaScript/TypeScript, SQL, C++
- **AI/ML:** LangChain, LangGraph, MCP, Triton Inference Server, RAG, PPO/TD3/DDPG, Anomaly Detection
- **Infrastructure:** Docker, Kubernetes, AWS, PostgreSQL/pgvector, Redis, Nginx
- **Frontend:** React, React Native, TypeScript
- **Observability:** Langfuse, SigNoz, custom metrics dashboards

Certificates & Languages

- **IELTS:** Overall 7.0 (C1) – Listening 7.5, Speaking 7.5, Reading 7.0, Writing 6.0 (09/2024)
- **Languages:** English (C2), German (A2), Turkish (Native)